

Proof of the Phase Kickback Phenomenon for the CNOT Gate

Optional, advanced. This note goes with Lab 3 and is not needed to complete the lab. It assumes the ket notation of the labs, the gates H , X and CNOT, and the tensor product $\otimes$ of two one-qubit states or gates. In $|ab\rangle = |a\rangle \otimes |b\rangle$ the first symbol a is the first qubit (here the control), as in the labs.

Introduction

In this short document we show in detail how the circuit

$$\text{CNOT}(H \otimes HX)$$

applied to the initial state $|00\rangle$ implements the transformation

$$|+-\rangle \xrightarrow{\text{CNOT}} |--\rangle,$$

where the first qubit is the control and the second is the target. This effect, in which the phase of the target is “kicked back” onto the control, is known as *phase kickback*.

Preliminaries

We recall the definitions of the single-qubit $|\pm\rangle$ states:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \quad (1)$$

The Hadamard gate H and Pauli X gate act on the computational basis as

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad (2)$$

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle. \quad (3)$$

Moreover, $|-\rangle$ is an eigenvector of X with eigenvalue -1 :

$$X|-\rangle = X\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -\frac{|0\rangle - |1\rangle}{\sqrt{2}} = -|-\rangle. \quad (4)$$

The CNOT gate with the first qubit as control and the second as target acts as

$$\text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle, \quad \text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle. \quad (5)$$

Preparing the state $|+-\rangle$

We start from the initial state $|00\rangle$ and apply the gates indicated in the circuit.

First qubit (control)

On the first qubit we apply a Hadamard gate:

$$(H \otimes I)|00\rangle = H|0\rangle \otimes |0\rangle = |+\rangle|0\rangle. \quad (6)$$

Second qubit (target)

On the second qubit we first apply X and then H , that is the one-qubit gate HX (X acts first):

$$X|0\rangle = |1\rangle, \quad HX|0\rangle = H|1\rangle = |-\rangle. \quad (7)$$

Overall, the two-qubit gate $H \otimes HX$ acts on each qubit separately, and after the local gates we obtain the state

$$|\psi\rangle = (H \otimes HX)|00\rangle = H|0\rangle \otimes HX|0\rangle = |+\rangle|-\rangle. \quad (8)$$

Let us now write $|\psi\rangle$ explicitly in the computational basis:

$$|\psi\rangle = |+\rangle|-\rangle \quad (9)$$

$$= \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \quad (10)$$

$$= \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle). \quad (11)$$

Action of the CNOT

We now apply the CNOT gate to the state $|\psi\rangle$ in (11):

$$|\psi'\rangle = \text{CNOT}|\psi\rangle \quad (12)$$

$$= \frac{1}{2}(\text{CNOT}|00\rangle - \text{CNOT}|01\rangle + \text{CNOT}|10\rangle - \text{CNOT}|11\rangle) \quad (13)$$

$$= \frac{1}{2}(|00\rangle - |01\rangle + |11\rangle - |10\rangle). \quad (14)$$

We group the terms according to the first qubit:

$$|\psi'\rangle = \frac{1}{2} \left[|0\rangle(|0\rangle - |1\rangle) + |1\rangle(|1\rangle - |0\rangle) \right]. \quad (15)$$

Note that

$$|0\rangle - |1\rangle = \sqrt{2}|-\rangle, \quad |1\rangle - |0\rangle = -\sqrt{2}|-\rangle, \quad (16)$$

and therefore

$$|\psi'\rangle = \frac{1}{2} \left[|0\rangle(\sqrt{2}|-\rangle) + |1\rangle(-\sqrt{2}|-\rangle) \right] \quad (17)$$

$$= \frac{\sqrt{2}}{2}(|0\rangle - |1\rangle)|-\rangle \quad (18)$$

$$= \frac{\sqrt{2}}{2}(\sqrt{2}|-\rangle)|-\rangle \quad (19)$$

$$= |-\rangle|-\rangle = |--\rangle. \quad (20)$$

We have thus shown explicitly that

$$\text{CNOT}|+-\rangle = |--\rangle. \quad (21)$$

Interpretation as phase kickback

From the point of view of the second qubit, we see that it remains in the state $|-\rangle$ both before and after the CNOT. However, from (4), the state $|-\rangle$ is an eigenstate of X with eigenvalue -1 . This means that when the control is $|1\rangle$, the action of CNOT is equivalent to multiplying the target state by a phase factor -1 .

Since the target is already in an eigenstate of X , the net effect is not a change of its state, but rather a *shift* of this phase onto the control qubit. In other words, the $|1\rangle$ component of the control acquires a minus sign with respect to its $|0\rangle$ component, turning $|+\rangle$ into $|-\rangle$. This transfer of phase from the target to the control is the *phase kickback* phenomenon.